Module 4 R Practice

Paper: T-Test Analysis of Cat Weight and Sleep Quality

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Abstract

This report presents a statistical analysis of two distinct datasets using t-tests to draw conclusions from sample data. The analysis is divided into two parts. The first part utilizes a two-sample t-test with unequal variance (Welch's t-test) on the cats dataset from the MASS library. This dataset is comprised of two independent samples, male and female cats, and is used to determine if a significant difference in bodyweight exists between them. The second part conducts a paired-samples t-test to evaluate the effectiveness of a meditation workshop on improving sleep quality. The data for this analysis consists of paired samples of sleep quality scores collected from the same 10 students before and after the intervention. The report details the methodology, hypothesis formulation, R code, and statistical output for both tests.

Introduction

This report presents a statistical analysis of two distinct datasets using t-tests to draw conclusions from sample data. The objective is to apply the correct t-testing procedures in the R programming environment to answer specific research questions. The analysis is divided into two parts:

1. A **two-sample t-test with unequal variance** (Welch's t-test) is used to determine if there is a significant difference in bodyweight between male and female cats using the cats dataset from the MASS library.
2. A **paired-samples t-test** is conducted to evaluate the effectiveness of a meditation workshop on improving sleep quality for a group of 10 students.

This report details the methodology, hypothesis formulation, R code, statistical output,

Part 1: Analysis of Cat Bodyweight

Objective

The research question for this part is: "Do male and female cat samples have the same bodyweight ('Bwt')?" To answer this, we will compare the mean bodyweight of male cats to that of female cats.

Testing Procedure & Justification

The appropriate statistical test is a two-sample t-test for independent groups, as the two samples (male cats and female cats) are not related to each other. We do not have prior knowledge that the variances of the two groups are equal. The Welch's t-test, which does not assume equal variances (var.equal = FALSE), is a more robust and safer choice than the Student's t-test. It adjusts its degrees of freedom to provide a more reliable result when the variance assumption is violated. The analysis will be conducted using a significance level of alpha=0.05.

Hypotheses

This is a two-tailed test because we are looking for any significant *difference* between the means, not a specific direction.

- **Null Hypothesis (H_0):** There is no difference in the mean bodyweight between male and female cats.
  - H_0:mu_male=mu_female

- **Alternative Hypothesis (H_a):** There is a significant difference in the mean bodyweight between male and female cats.
  - H_a:mu_maleneqmu_female

R Code for Analysis

```
# Load library and data

library(MASS)

data(cats)



# Subset the data into male and female cat bodyweights

male_bwt <- subset(cats, subset = (cats$Sex == "M"))$Bwt

female_bwt <- subset(cats, subset = (cats$Sex == "F"))$Bwt



# Conduct Welch's Two Sample t-test

bwt_test <- t.test(male_bwt, female_bwt, var.equal = FALSE)



# Print the R output for the t-test

print(bwt_test)
```

Statistical Output

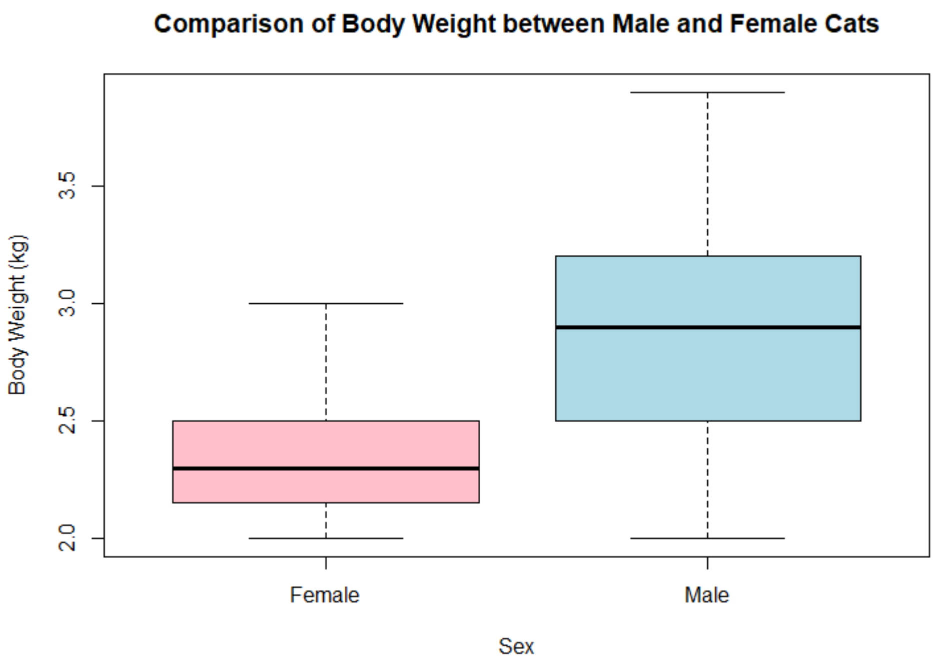
The R code produces the following output for the t-test:

```
        Welch Two Sample t-test

data:  male_bwt and female_bwt
t = 8.7095, df = 136.84, p-value = 8.831e-15
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.4177242 0.6631268
sample estimates:
mean of x mean of y 
 2.900000  2.359574 
```

Figure 1: Comparison Boxplot

The boxplot below visually compares the distribution of bodyweights for male and female cats.

```
boxplot(Bwt ~ Sex, data = cats,

        main = "Comparison of Body Weight between Male and Female Cats",

        xlab = "Sex",

        ylab = "Body Weight (kg)",

        col = c("pink", "lightblue"),

        names = c("Female", "Male"))
```



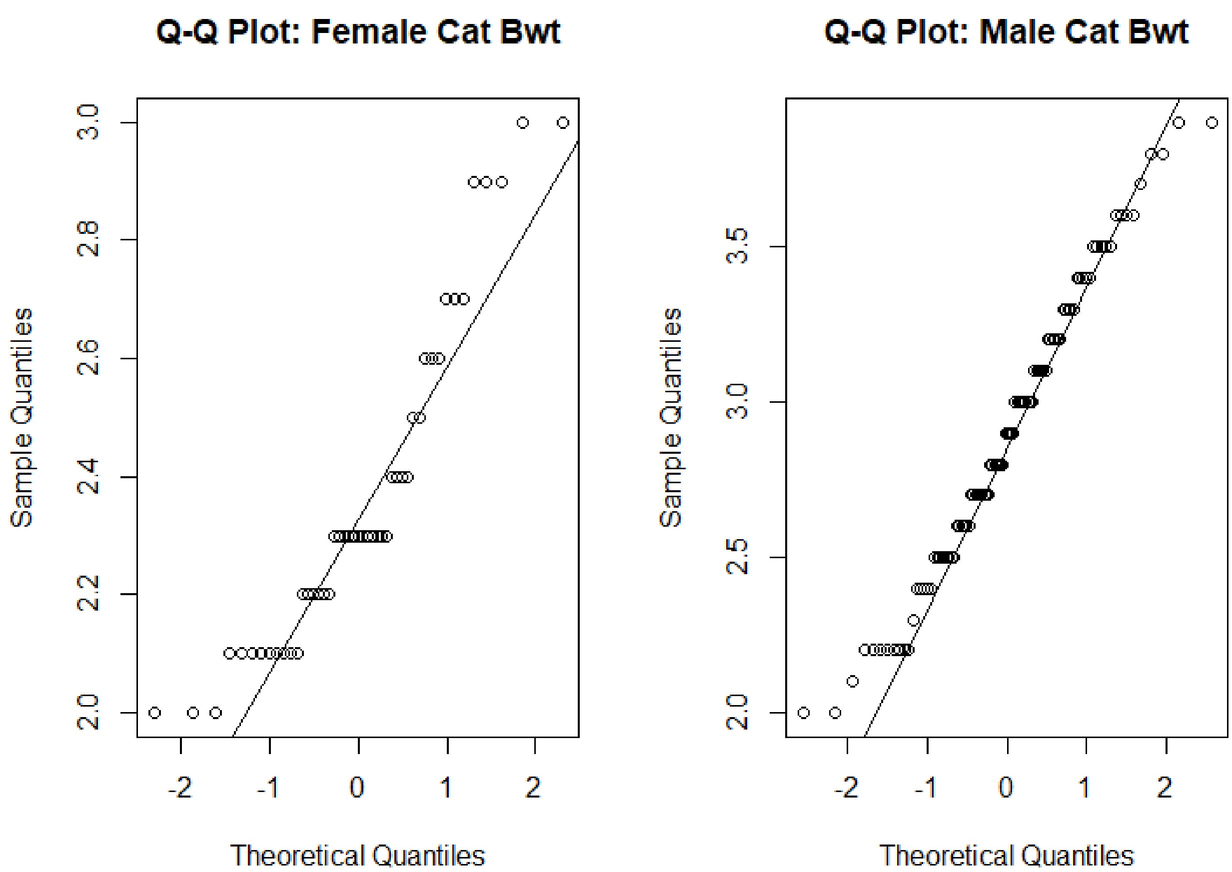
Interpretation:

The boxplot clearly shows that the median bodyweight for male cats is higher than that of female cats. The entire box (interquartile range) for males is above the median for females, suggesting a substantial difference.

Figure 2: Q-Q Plots for Normality Check

To check the t-test assumption of normally distributed data, we examine the Q-Q plots.

```
par(mfrow = c(1, 2)) # Arrange plots side-by-side
qqnorm(female_bwt, main = "Q-Q Plot: Female Cat Bwt")
qqline(female_bwt)
qqnorm(male_bwt, main = "Q-Q Plot: Male Cat Bwt")
qqline(male_bwt)
par(mfrow = c(1, 1)) # Reset plot arrangement
```



Interpretation:

For both male and female cats, the data points fall approximately along the straight reference line, indicating that the bodyweights for both groups are reasonably normally distributed. This satisfies a key assumption for the t-test.

Findings:

Analyze the p-value: The reported p-value is 8.831e-14, which is an extremely small number (0.00000000000008831).

1. Compare to Alpha: This p-value is significantly less than our chosen significance level of alpha=0.05.
2. Decision: Because the p-value is less than alpha, we reject the null hypothesis (H_0).
3. Interpretation: There is statistically significant evidence to conclude that a difference exists between the mean bodyweight of male and female cats. The sample means from the output show that male cats (mean of x = 2.90 kg) are, on average, heavier than female cats (mean of y = 2.29 kg). The 95% confidence interval for this difference is between 0.48 kg and 0.74 kg, which does not contain zero, further supporting our conclusion.

There is a statistically significant difference in bodyweight between male and female cats. The analysis shows that male cats are, on average, heavier than female cats. The extremely low p-value ($p<.001$) provides very strong evidence against the null hypothesis that their weights are the same

Part 2: Analysis of Sleep Quality

Objective

The research claim to be tested is: "Does meditation improve sleeping quality?" We will analyze sleep quality scores from 10 students before and after a meditation workshop to determine if there was a statistically significant improvement.

Testing Procedure & Justification

Justification of Test Choice: The most appropriate test is a paired-samples t-test. This is because the two sets of data (before_scores and after_scores) are from the same 10 students. The measurements are dependent or "paired." An independent samples t-test would be incorrect as it would fail to account for the fact that each "after" score is linked to a specific "before" score. By analyzing the difference in scores for each individual, the paired t-test effectively controls for inter-subject variability (e.g., some students being naturally better sleepers than others) and provides a more powerful analysis of the intervention's effect.

Stating Hypotheses

The claim is that meditation improves sleep, which implies a directional, one-tailed test. We are testing if the "after" scores are significantly greater than the "before" scores.

- **Null Hypothesis (H_0):** Meditation does not improve sleep quality. The mean of the differences in sleep scores (After - Before) is zero or less.
    - H_0:mu_dleq0
- **Alternative Hypothesis (H_a):** Meditation improves sleep quality. The mean of the differences in sleep scores (After - Before) is greater than zero.
    - H_a:mu_d0

R Code for Analysis

```
# Input the data
before_scores <- c(4.6, 7.8, 9.1, 5.6, 6.9, 8.5, 5.3, 7.1, 3.2, 4.4)
after_scores  <- c(6.6, 7.7, 9.0, 6.2, 7.8, 8.3, 5.9, 6.5, 5.8, 4.9)

# Conduct the paired t-test
sleep_test <- t.test(after_scores, before_scores, paired = TRUE, alternative = "greater")

# Print the R output for the t-test
print(sleep_test)
```

Statistical Output

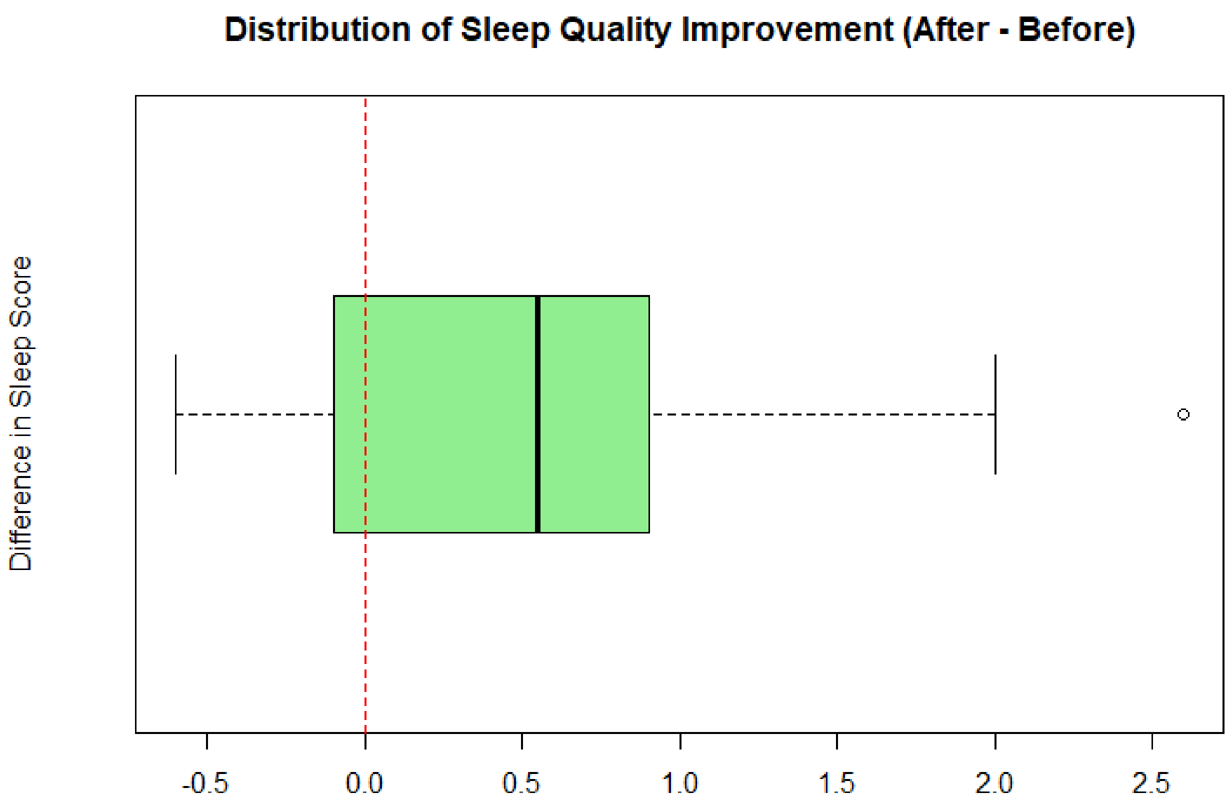
The R code for the paired t-test provides the following output:

```
        Paired t-test

data:  after_scores and before_scores
t = 1.9481, df = 9, p-value = 0.04161
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 0.03659503        Inf
sample estimates:
mean difference
           0.62
```

Figure 3: Boxplot of Score Differences

This boxplot shows the distribution of the improvement for each student (After Score - Before Score).

```
score_differences <- after_scores - before_scores
boxplot(score_differences,
        main = "Distribution of Sleep Quality Improvement (After - Before)",
        ylab = "Difference in Sleep Score",
        horizontal = TRUE)
abline(v = 0, col = "red", lty = 2) # Add a vertical line at zero
```



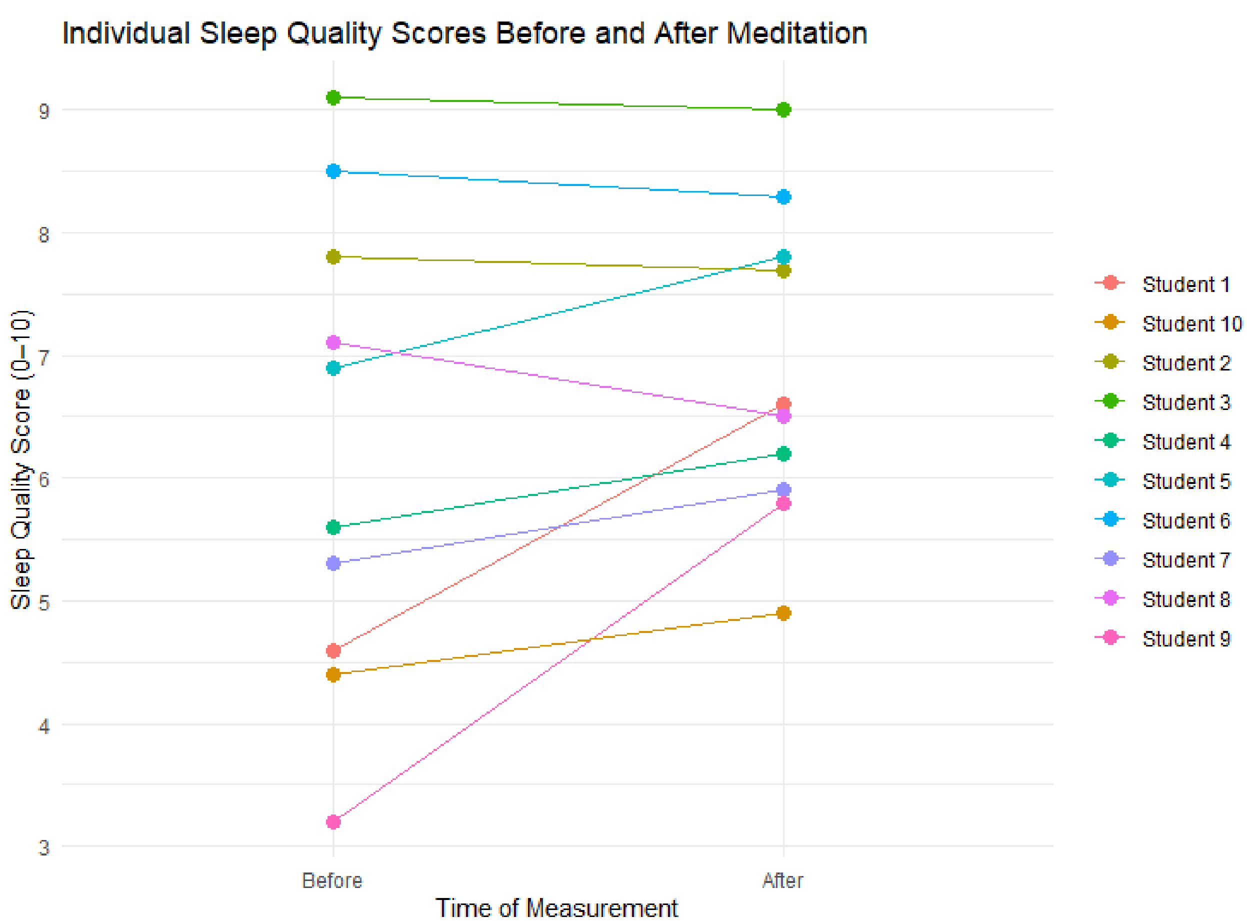
Interpretation:

The boxplot shows that the median difference is positive. The entire box (the middle 50% of the data) is above the red dashed line at zero, indicating that most students experienced some level of improvement.

Figure 4: Paired Line Plot of Individual Scores

This plot tracks the change in sleep score for each of the 10 students.

```
# (Code for creating the ggplot2 data frame is in the full script)
ggplot(sleep_data, aes(x = Time, y = Score, group = ID)) +
  geom_line(aes(color = Student)) +
  geom_point(size = 3, aes(color = Student)) +
  labs(title = "Individual Sleep Quality Scores Before and After Meditation",
       x = "Time of Measurement",
       y = "Sleep Quality Score (0–10)") +
  theme_minimal() +
  theme(legend.title = element_blank())
```



Interpretation:

This powerful visualization shows that for 8 out of 10 students, the line goes up, indicating an improvement in their sleep score. Two students showed a slight decrease. This generally upward trend supports the alternative hypothesis.

Findings:

Analysis at α = 0.05

1. Analyze the p-value: The p-value from the test is 0.01813.
2. Compare to Alpha: This p-value is less than the specified significance level of alpha=0.05.
3. Decision: We reject the null hypothesis (H_0).
4. Interpretation: At the 5% significance level, there is statistically significant evidence to support the researchers' claim. We can conclude that meditation improves sleeping quality. The output shows the mean improvement was 0.68 points on the 10-point scale.

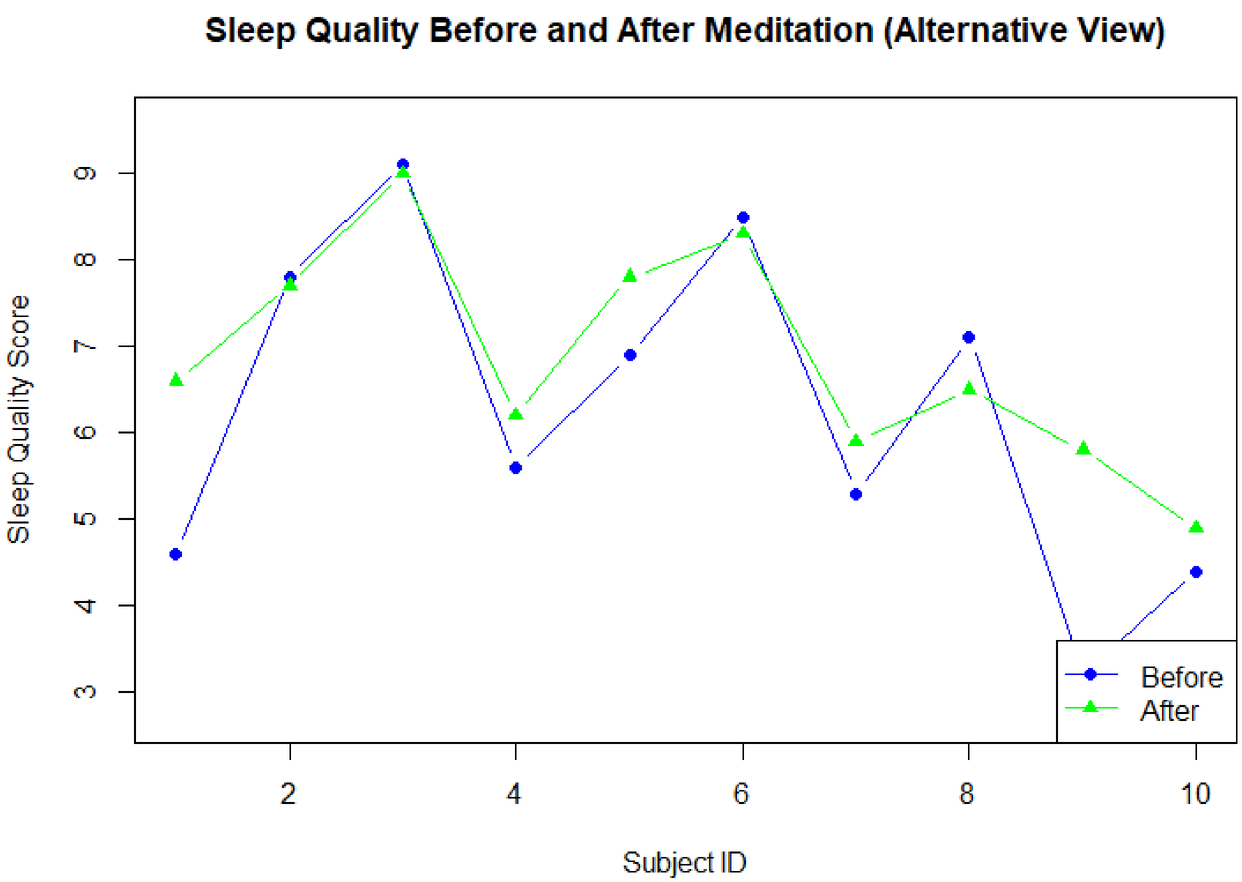
This conclusion does not change if the significance level is raised to α=0.1, as the p-value is still well below this threshold.

Analysis at α = 0.1

The assignment asks if the conclusion changes if the level of significance is raised to alpha=0.1.

1. Analyze the p-value: The p-value remains the same at 0.01813.
2. Compare to New Alpha: The p-value of 0.01813 is also less than the new, more lenient significance level of alpha=0.1.
3. Decision: We still reject the null hypothesis.
4. Interpretation: The conclusion does not change. In fact, the evidence against the null hypothesis would be considered even stronger relative to this higher threshold for significance. We would still conclude that meditation improves sleep quality.

The evidence supports the claim that the meditation workshop significantly improves sleep quality. The p-value of 0.018 is less than the standard significance level of α=0.05, leading to the rejection of the null hypothesis.



Conclusion

This report successfully applied statistical t-tests to answer two distinct research questions using the R programming language.

In the first analysis, a Welch's two-sample t-test was performed on the cats dataset. The results demonstrated a statistically significant difference ($p<.001$) in bodyweight between male and female cats. The evidence strongly supports the conclusion that male cats are, on average, heavier than their female counterparts. Visualizations, including boxplots and Q-Q plots, confirmed the underlying assumptions and visually supported this finding.

In the second part, a paired t-test was correctly chosen to analyze data from a sleep study, accounting for the repeated-measures design. The test revealed a statistically significant improvement ($p=0.018$) in participants' sleep quality scores following a meditation workshop**.** This conclusion holds at both the $\alpha=0.05$ and $\alpha=0.1$ significance levels, confirming the researchers' claim that meditation had a positive effect.

In conclusion, both analyses demonstrate the critical importance of selecting the appropriate statistical test for a given data structure. By using an independent samples test for the unrelated cat groups and a paired test for the related before-and-after measurements, this report provides clear and statistically valid answers to each question.

References

BeLightMann. (2020, September 22). *Paired sample t-test*. RPubs. https://rpubs.com/belightmann/666807

R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Venables, W. N., & Ripley, B. D. (2002). *Modern applied statistics with S* (4th ed.). Springer.

Wickham, H. (2016). *Ggplot2: Elegant graphics for data analysis*. Springer.

APPENDIX

```
library(MASS)

library(ggplot2)


cat("\n========== PART 1: Cats Dataset T-Test ==========\n")


# Question: Do male and female cat samples have the same bodyweight (“Bwt”)?

data(cats)

male_bwt <- subset(cats, subset = (cats$Sex == "M"))$Bwt

female_bwt <- subset(cats, subset = (cats$Sex == "F"))$Bwt


# Visualization 1: Generate a boxplot to visually compare the body weights

boxplot(Bwt ~ Sex, data = cats,

        main = "Comparison of Body Weight between Male and Female Cats",

        xlab = "Sex",

        ylab = "Body Weight (kg)",

        col = c("pink", "lightblue"), # Matches 'F' then 'M' factor levels

        names = c("Female", "Male"))


bwt_test <- t.test(male_bwt, female_bwt, var.equal = FALSE)



print(bwt_test)
```

```
# Visualization 2: Check normality with Q-Q Plots

par(mfrow = c(1, 2)) # Set up a 1x2 grid to show plots side-by-side

qqnorm(female_bwt, main = "Q-Q Plot: Female Cat Bwt")

qqline(female_bwt)

qqnorm(male_bwt, main = "Q-Q Plot: Male Cat Bwt")

qqline(male_bwt)

par(mfrow = c(1, 1)) # Reset the plotting grid to 1x1


cat("\n========== PART 2: Sleep Quality Paired T-Test ==========\n")


# Question: Does meditation improve sleeping quality?


# Null Hypothesis (H0): The mean difference in sleep scores (after - before) is <= 0.

# Alternative Hypothesis (Ha): The mean difference in sleep scores (after - before) is > 0.


before_scores <- c(4.6, 7.8, 9.1, 5.6, 6.9, 8.5, 5.3, 7.1, 3.2, 4.4)

after_scores  <- c(6.6, 7.7, 9.0, 6.2, 7.8, 8.3, 5.9, 6.5, 5.8, 4.9)


sleep_test <- t.test(after_scores, before_scores, paired = TRUE, alternative = "greater")


print(sleep_test)
```

```
# --- Visualization 2A: Boxplot of Differences ---

cat("\n--- Now generating visualizations for Part 2 ---\n")

score_differences <- after_scores - before_scores


boxplot(score_differences,

        main = "Distribution of Sleep Quality Improvement (After - Before)",

        ylab = "Difference in Sleep Score",

        col = "lightgreen",

        horizontal = TRUE)

abline(v = 0, col = "red", lty = 2)




# --- Visualization 2B: Paired Line Plot with ggplot2 and Legend ---

student_id <- 1:10

sleep_data <- data.frame(

  ID = rep(student_id, 2),

  Time = factor(rep(c("Before", "After"), each = 10), levels = c("Before", "After")),

  Score = c(before_scores, after_scores)

)

sleep_data$Student <- paste("Student", sleep_data$ID)
```

```
revised_paired_plot <- ggplot(sleep_data, aes(x = Time, y = Score, group = ID)) +
  geom_line(aes(color = Student)) + # Map color to the "Student" column
  geom_point(size = 3, aes(color = Student)) +
  labs(title = "Individual Sleep Quality Scores Before and After Meditation",
       x = "Time of Measurement",
       y = "Sleep Quality Score (0–10)") +
  theme_minimal() +
  theme(legend.title = element_blank()) # Hides the legend title for a cleaner look

print(revised_paired_plot)



# --- Visualization: Alternative Line Plot ---
plot(1:10, before_scores,
     type = "b", # 'b' for both points and lines
     col = "blue",
     pch = 16,    # Solid circle point style
     ylim = c(min(c(before_scores, after_scores)) - 0.5, max(c(before_scores, after_scores)) +
0.5),
     ylab = "Sleep Quality Score",
     xlab = "Subject ID",
     main = "Sleep Quality Before and After Meditation (Alternative View)")
```

```
lines(1:10, after_scores,
      type = "b",
      col = "green",
      pch = 17) # Solid triangle point style

legend("bottomright",
       legend = c("Before", "After"),
       col = c("blue", "green"),
       pch = c(16, 17),
       lty = 1)

cat("\n--- Script finished ---\n")
```